

Planetary Defense

Astronomy · AP Physics 1 mechanics

A mountain ridge: one road, domes with open shutters, a thirty-metre dish and red service lamps.

45	14	5	6/10	1	14 · 30 · 1
STOPS	CHOICE (31%)	EFFECTIVE FORMATS	EQUATIONS COMPUTED	SELECT-ONLY CONCEPTS	SELECT · CONSTRUCT · OPERATE

READING Where this campaign stands

Five gaps, three of them on instruments and likely Phase 0 phantoms. Only one select-only concept, which is the best figure in the set. Fourteen CHOICE stops is the weakness. The place is nocturnal and linear — one dark road along a ridge, domes with open shutters, a thirty-metre dish and red service lamps — which suits the formats that run on their own clock, because an observing night is already a closing window.

NOW The mix as it ships

CHOICE 14	BALLPARK 11	PROTOCOL 8	SEQUENCE 4	VALUE 1	CLOUD 1	DEGENERACY 1	TRACE 1	INJECT 1	VERIFY 1	TRIGGER 1	ATTEST 1
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CURRICULUM What the syllabus is not getting

EQUATION THE COURSE OWES	WHAT IT CARRIES	MENTIONED AT	STATUS
$d = \text{baseline} / \text{parallax angle}$	distance from a parallax shift	5 · SEQUENCE 6 · VALUE 8 · PROTOCOL 9 · CHOICE	verify first
$v = \sqrt{GM / r}$	orbital speed, and the energy that goes with it	25 · BALLPARK 38 · PROTOCOL 39 · CHOICE	real gap
$\Delta\lambda / \lambda = v / c$	Doppler shift as a line-of-sight speed	8 · PROTOCOL 22 · PROTOCOL	real gap
$\lambda_{\text{max}} T = b$	Wien's law — a temperature read off a thermal spectrum	15 · DEGENERACY	verify first

2 of 4 sit on an instrument stop, whose board arithmetic the coverage check cannot yet read — expect Phase 0 to clear those. 2 are unaffected by it.

TIER Concepts the player only ever picks from a list

#	MECHANISM CONCEPT	STOPS
30	Crater and blast scaling; ground effects	29

Reported, never failed. Select-tier can be exactly right; these are the candidates, not the defects. The syllabus also lists 1 concept no stop touches at all — **Planetary perturbations and resonances** — which is expected, and is a different conversation from this one.

TALLY STRONG

stop 5 or 9 · ORBIT / OPS

Astrometric observations accumulate and the orbit tightens; early on the arc moves because there is not enough of it, not because the object did anything. "Build the observation arc" is currently a SEQUENCE, and this is the same idea with the player deciding when to stop.

SERVES the parallax and orbit-determination gaps

NEEDS an early arc that flatters the solution

SWEEP STRONG

stop 15 · CHAR

A thermal spectrum has a peak, and Wien's law is where it sits. The game already has a DEGENERACY here about bounding the diameter; a SWEEP over wavelength is the measurement that breaks the degeneracy, and it is plotted only where you look.

SERVES $\lambda_{\text{max}} T = b$

NEEDS the peak unmarked

TRIAL STRONG

the ridge road

One road, played entirely at night, 1.6 km of it with base camp in the first 200 m. A driven order between domes set by what is above the horizon when — the sky is the constraint and it moves.

SERVES observing planning; the nocturnal ridge made load-bearing

RISK world work, and the site already has mapRadius set for exactly this scale

HOLD STRONG

RADAR · the dish

Hold a thirty-metre dish on target while wind and tracking error push it, inside a band that closes as the object gets faster. Physical, real, and the failure is losing the pass.

SERVES tracking, and angular rates

BELT STRONG

DISC · Survey Telescope

A survey pipeline sorts detections into real and artefact, at speed, all night. That is exactly one bit of subject matter held under pressure, and the game already carries an INJECT about pushing a known population through the pipeline — so the BELT is the pipeline the INJECT measures.

SERVES detection and false positives; pairs with the existing INJECT

RISK decidable at a glance, or it is dexterity

SPOT STRONG

an OPS stop

An observing priority replaced mid-night. A ridge full of red lamps and a target setting in forty minutes is the most natural home for an instruction that changes without ceremony.

SERVES triage of observing time

PROBE MEDIUM

The ridge is long enough to walk a chain along, but there is no physical quantity that varies down it. Medium at best.

LOB REJECT

A kinetic impactor is the obvious hook and the wrong one: the whole point of deflection is that a centimetre a second is **computed** years ahead. The game says so at stop 39.

SLATES **Three appetites — the campaign is zero-sum**

No conversion adds a stop. Each changes what an existing stop **does** while leaving what it **teaches** alone. Snapshot before, diff after: curriculumDelivery.mjs planetary_defense --snapshot before.json , then --against before.json .

Minimum honest 3 STOPS · ONE SITTING

9	d3	OPS	CHOICE → TALLY	Collapse the orbit uncertainty the orbit collapsed by accumulating arc, not by picking
15	d5	CHAR	DEGENERACY → SWEEP	Bound the diameter the diameter bound broken by a spectrum you scan
29	d10	OPS	CHOICE → CHAIN	Why did the object break up high in the atmosphere? why it broke up high — name the governing transfer

CHOICE **14 → 12** (27%) · moves **14·30·1 → 12·30·3** · effective formats **5.0 → 5.7** · **2** new operate stops

Balanced 6 STOPS · THE RECOMMENDATION

9	d3	OPS	CHOICE → TALLY	Collapse the orbit uncertainty
15	d5	CHAR	DEGENERACY → SWEEP	Bound the diameter
29	d10	OPS	CHOICE → CHAIN	Why did the object break up high in the atmosphere?
39	d13	OPS	CHOICE → BALLPARK	Why a centimetre a second is enough why a centimetre a second is enough — computed
22	d8	RADAR	PROTOCOL → HOLD	What does radar measure? the dish held on target through the pass
26	d9	OPS	BALLPARK → CLOUD	The speed it actually arrives at the arrival speed as a band the measurements permit

CHOICE **14 → 11** (24%) · moves **14·30·1 → 11·30·4** · effective formats **5.0 → 6.3** · **3** new operate stops, 1 of them arcade

Ambitious 9 STOPS · ADDS WORLD WORK

9	d3	OPS	CHOICE → TALLY	Collapse the orbit uncertainty
15	d5	CHAR	DEGENERACY → SWEEP	Bound the diameter
29	d10	OPS	CHOICE → CHAIN	Why did the object break up high in the atmosphere?
39	d13	OPS	CHOICE → BALLPARK	Why a centimetre a second is enough
22	d8	RADAR	PROTOCOL → HOLD	What does radar measure?
26	d9	OPS	BALLPARK → CLOUD	The speed it actually arrives at
8	d3	OPS	PROTOCOL → BELT	Which orbit feature does the data constrain? the survey pipeline, sorted at the speed detections arrive
38	d13	OPS	PROTOCOL → SPOT	When does it come back? an observing priority replaced mid-night
5	d2	ORBIT	SEQUENCE → TRIAL	Build the observation arc the ridge road, driven, against a sky that moves

CHOICE **14 → 11** (24%) · moves **14·30·1 → 11·27·7** · effective formats **5.0 → 6.9** · **6** new operate stops, 4 of them arcade

THE ORDER OF WORK

1. Phase 0 — teach the coverage check to read an instrument's own board, and re-baseline `curriculum-debt.json`. Rewriting a stop that already teaches the thing is the exact failure the check exists to prevent. **2.** Snapshot. **3.** Edit the book, never the generated content — `tools/import-book.mjs` `books/planetary-defense.yml` `planetary_defense --verify`. **4.** `npm run traps`: every instrument's trap must fire. **5.** `npm run check planetary_defense` and `npm run drive planetary_defense`, right answer and wrong — ask what a player who understood nothing would score before believing any pass mark. **6.** `--against the snapshot`. A changed format is reported; a changed objective fails.

Counts computed from `themes/planetary_defense/content/` via `engine/dev/curriculumDelivery.mjs`; concept and equation lists authored in `tools/syllabus.js`. Format verdicts and slates are editorial. Run it with `THEME=planetary_defense npm run dev`.